8.5 Partial Derivatives

In this section, we shall see how the concept of derivative for functions of one variable may be generalized to real-valued function of several variables. First we consider functions of two variables. Let $A=\{(x,y)|a<x<b, c<y<d\}\subset\mathbb{R}^2$, and $F:A\to\mathbb{R}$ be a real-valued function. Suppose that $(x_0,y_0)\in A$; and we are interested in finding the rate of change of F at (x_0,y_0) with respect to the change **only** in the variable x. As we have seen above $F(x,y_0)$ is a function of x alone and it will represent a curve obtained by intersecting the surface $z=F(x,y)$ with $y=y_0$ plane. So we can discuss the slope of the tangent to the curve $z=F(x,y_0)$ at $x=x_0$ by finding derivative of $F(x,y_0)$ with respect to x and evaluating it at $x=x_0$. Similarly, we can find the slope of the curve $z=F(x_0,y)$ at $y=y_0$ by finding derivative of $F(x_0,y)$ with respect to y and evaluating it at $y=y_0$. These are the key ideas that motivate us to define partial derivatives below.

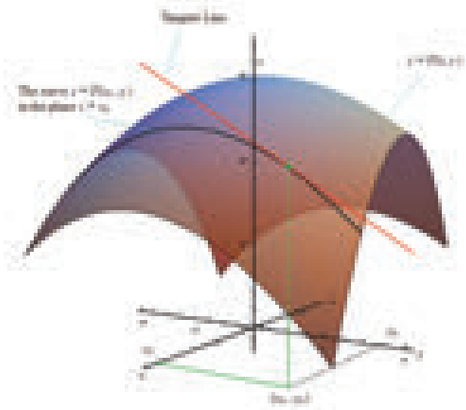


Fig. 8.11

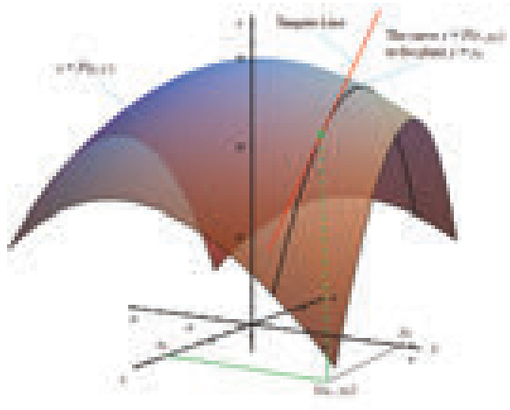


Fig. 8.12

Definition 8.8

Let $A=\{(x,y)|a<x<b, c<y<d\}\subset\mathbb{R}^2, F:A\to\mathbb{R}$ and $(x_0,y_0)\in A$.

(i) We say that F has a partial derivative with respect to x at $(x_0,y_0)\in A$ if

$$\lim_{h\to 0}\frac{F(x_0+h,y_0)-F(x_0,y_0)}{h} \qquad \ldots (10)$$

exists. In this case, the limit value is denoted by $\frac{\partial F}{\partial x}(x_0,y_0)$.

(ii) We say F has a partial derivative with respect to y at $(x_0,y_0)\in A$ if

$$\lim_{k\to 0}\frac{F(x_0,y_0+k)-F(x_0,y_0)}{k} \qquad \ldots (11)$$

exists. In this case, the limit value is denoted by $\frac{\partial F}{\partial y}(x_0,y_0)$.

Remarks

1. Partial derivatives for functions of three or more variables are defined exactly in a similar manner.

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